

Layering and double-diffusion style convection in Europa's ocean

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Received 20 July 2004; revised 3 June 2005

Available online 8 August 2005

Abstract

We assess the effects of fluid composition and depth-dependent hydrostatic pressure on dynamics in a euroman ocean primarily composed of aqueous Na_2SO_4 and MgSO_4 , for both a saturated and a dilute ocean. We observe a salinity- and pressure-dependent check on buoyancy in putative upwellings, which could act as a mechanism for storing heat in the ocean's base. Uptake of salt to a warmed parcel of water from the seafloor environment causes upwellings to lose buoyancy before reaching the base of the overlying ice, implying a tendency toward ocean stratification. We make an analogy to double-diffusive convecting systems observed in the Red Sea and elsewhere in Earth's waters. Using currently understood parameterizations for onset and stability, we estimate a range of temperatures and salinities for which double-diffusive convection may occur in Europa's ocean. In the Red Sea, boundary layers separating convecting zones have been observed to move upward as the lower layer acquires heat and salt. We examine the possibility of stratification and double-diffusive convection as mechanisms for heat storage in Europa's ocean, and possible correlation with recently inferred changes in ice surface alteration style over the last 30–80 Myr. © 2005 Elsevier Inc. All rights reserved.

Keywords: Europa; Interiors; Thermal history; Stratification

1. Introduction

Europa's ocean has received much attention (Kargel et al., 2000; Collins et al., 2000; Thomson and Delaney, 2001; Goodman et al., 2004; Melosh et al., 2004) since the Galileo mission revealed evidence for a subsurface water layer (Zimmer et al., 2000). Interpretations of Galileo data provide constraints on the surface ice thickness and depth of the underlying ocean (Anderson et al., 1998; Hoppa et al., 1999; Turtle and Pierazzo, 2001). Non-ice material on the surface appears to be representative of ocean composition (McCord et al., 1998). Spectral resemblance of the non-ice material to sodium and magnesium sulfates is consistent with a chondritic origin of the moon (Kargel, 1991; Kargel et al., 2000). Mass balance calculations (Kargel et al., 2000; Zolotov and Shock, 2001) suggest a eutectic Na_2SO_4 and MgSO_4 ocean composition subject to alteration by interaction with Europa's rocky mantle. However, McKinnon

and Zolensky (2003) caution that chondrite composition and rock hydration chemistry are topics that remain poorly constrained. They suggest that a more dilute, sulfate-dominated ocean is equally feasible. Crater counts constrain Europa's surface age to 30–80 Myr (Zahnle et al., 2003). Stratigraphic analyses (Figueredo and Greeley, 2004) indicate that Europa's surface has undergone at least two distinct eras of surface alteration during its history; the most recent dominated by melt-through, or chaos, features.

Dynamics in the ocean have been described in terms heat input (radiogenic and tidal) to Europa, its overall size and rate of rotation, and the depth of its ocean (Collins et al., 2000; Thomson and Delaney, 2001; Goodman et al., 2004). Assuming the ocean salinity is homogeneous, Goodman et al. (2004) describe a buoyant plume originating from the floor of Europa's ocean. The plume remains buoyant over entire depth of the ocean and reaches the base of the ice shell as a rotationally confined column with diameter on the order of tens of kilometers, comparable to the size of observed melt-through features on the surface. Melosh et al. (2004) point out that, in a dilute ocean, a rising plume would lose

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buoyancy just before reaching the surface, due to water's anomalous thermal expansion properties (at low temperature and pressure, water has negative thermal expansion). Specifically, the effect would occur in the 23 km below Europa's surface, but only if the ice shell is thinner than 23 km and if the water is sufficiently dilute—salinity less than 3.4 wt% for seawater, figures that are only poorly constrained by limited data pertaining to low temperature, high pressure thermochemistry. To date, no one has considered the effect of a salinity anomaly in a rising plume.

Pressure at the base of a 70–180 km ocean (Anderson et al., 1998) is 100–200 MPa. Until recently (Marion et al., 2005), researchers modeling chemical and physical properties of Europa's ocean have treated solution properties only at 0.1 MPa (1 atm), largely due to a lack of experimental data at low temperature and high pressure (Kargel et al., 2000; Melosh et al., 2004). We emphasize that the literature pertaining to high pressure (Na₂Mg)SO₄ is indeed sparse (e.g., Hogenboom et al., 1995; Abramson et al., 2001). Our model calculations were performed using a model that extrapolates to high pressures and low temperatures.

Using a simple calculation for the density anomaly of a thermal upwelling (hereafter referred to as a plume) versus pressure, we show that a small uptake of salt at the seafloor causes an initially buoyant upwelling to lose buoyancy at a height dependent on temperature and salinity of the plume and of the ambient ocean.

2. The influence of pressure thermochemistry on buoyancy in Europa's ocean

We test the buoyancy of a plume at all depths in Europa's ocean by calculating the density anomaly, B , with plume buoyancy occurring when

$$B \equiv \frac{\rho_{\text{ocean}} - \rho_{\text{plume}}}{\rho_{\text{plume}}} > 0. \quad (1)$$

For the calculation of densities, we model volumes of aqueous solutions as described below.

2.1. Volumetric properties of aqueous solutions: theory and limitations

The total volume of a unit of solution containing one kg of water and i dissolved species can be described by

$$V_{\text{total}} = V_{\text{water}} + \sum_i n_i V_i^{\circ} + V^{\text{ex}}. \quad (2)$$

V_{water} is the volume of one kg of water at the chosen pressure and temperature. The term V_i° for the i th species is its partial molal volume at infinite dilution ($n_i = 0$),

$$V_i^{\circ} = \left(\frac{\partial V}{\partial n_i} \right)_{P,T,n_j \neq i}, \quad n_i = 0 \quad (3)$$

at a given pressure and temperature. V_i° may be thought of as the species' contribution, per n_i moles, to the volume of the system when the forces mediating its interaction with the water are taken into account. For many salts, V_i° is negative over a range of pressures and temperatures, indicating that the addition of salt decreases the volume of the solution. V^{ex} the excess volume, becomes important in more highly concentrated solutions, in which abundance of dissolved ions causes stronger ion–ion interactions.

Density at a given temperature and pressure is related to volume and mass by

$$\rho = \frac{M_{\text{total}}}{V_{\text{total}}} = \frac{1000 + \sum_i n_i W_i}{V_{\text{water}} + \sum_i n_i V_i^{\circ} + V^{\text{ex}}}. \quad (4)$$

W_i is the molar mass (g mol^{−1}) of the i th constituent. 1000 g is, by definition, the amount of water in a molal quantity, and V_{water} the volume of 1000 g of water at the chosen pressure and temperature.

For water, volumes are obtained from the IAPWS95 formulation (Wagner and Pruss, 2002). We use values of V_i° from Marion et al. (2005). We find that supcrt92 (Shock et al., 1992) values give better agreement with densities measured at pressures above 0.1 MPa (Hogenboom et al., 1995). However, a comparably good fit is obtained from Marion et al. (2005) by applying an empirical factor of 0.5 in the compressibility term in V_i° . Temperature effects on V_i° are responsible for the phenomena described below, and supcrt92 and Marion et al. (2005) give good agreement in temperature, with Marion et al. (2005) providing the smoother fit. Our density calculations differ from experimental densities (Hogenboom et al., 1995) by no more than 5% (Figs. 1 and 2), the precision also obtained by Marion et al. (2005).

2.2. Assumptions and methods

As pointed out by Melosh et al. (2004) an unstratified, fully convective european ocean is nearly isothermal. Adiabatic increase in temperature with depth happens because material is being compressed with no exchange of heat with the environment. It follows (cf. Turcotte and Schubert, 1982) that the incremental increase in temperature with depth is

$$\left(\frac{\partial T}{\partial z} \right)_s = \frac{\alpha_V g T}{C_p} \quad (5)$$

where z is depth, α is the coefficient of thermal expansion, for water $O(2 \times 10^{-4})$ K^{−1}. Europa's gravity is 1.31 m s^{−2}, increasing to roughly 1.5 m s^{−2} approaching the rocky mantle (Turcotte and Schubert, 1982). T is the temperature, in K, at the depth of concern. C_p is the heat capacity of the fluid at the pressure and temperature of concern, and is about 4 kJ kg^{−1} K^{−1}. The adiabatic increase in temperature is in the neighborhood of 0.002 K km^{−1}, or 0.2 K over the entire depth of a 100 km ocean.

For a rotationally confined plume with a thermal anomaly of 0.2–1 mK (Goodman et al., 2004), we assess plume buoyancy in a dilute MgSO₄ ocean and a saturated eutectic ocean.

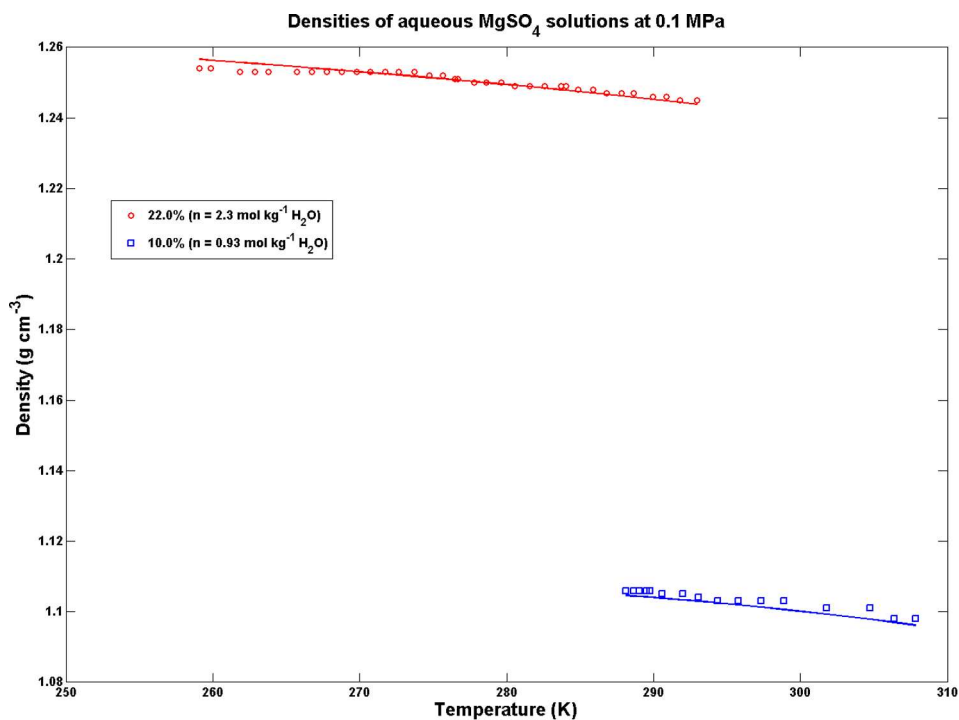


Fig. 1. Predictions of solution density (lines) at pressure up to 400 MPa, and 0.1 MPa in comparison with values measured by Hogenboom et al. (1995) (points, digitized from published data). Agreement is to within 5%. Predictions were made using the IAPWS95 equation of state for water (Wagner and Pruss, 2002). Molal volumes and excess volumes were calculated from Marion et al. (2005), with a factor of 0.5 in the compressibility term for the molal volume added following discussion with G. Marion.

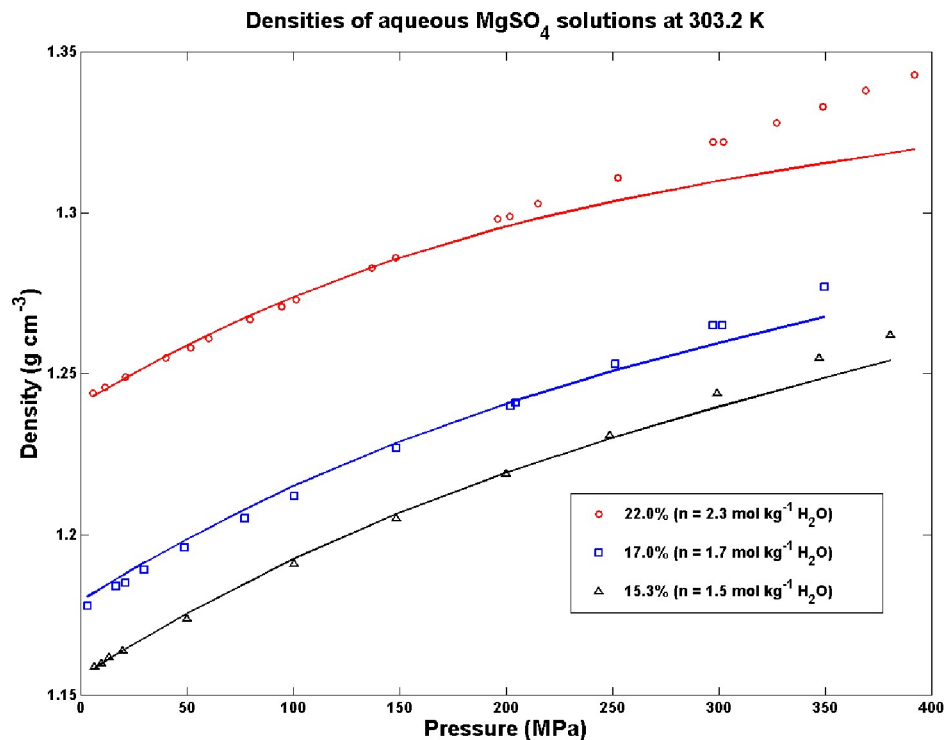


Fig. 2. Predictions of solution density (lines) at 303.2 K, and temperatures 260 to 310 K, in comparison with values measured by Hogenboom et al. (1995) (points, digitized from published data). Agreement is to within 5%. Predictions were made using the IAPWS95 equation of state for water (Wagner and Pruss, 2002). Molal volumes and excess volumes were calculated from Marion et al. (2005), with a factor of 0.5 in the compressibility term for the molal volume added following discussion with G. Marion.

Choosing a dilute Na_2SO_4 ocean instead, the ocean composition and plume enrichment for which a specific critical rise height is observed vary by no more than 10%, respectively. For the saturated ocean, we estimate the saline enrichment of the plume from Marion et al.'s (2005) saturation profile in Mg^{2+} , Ca^{2+} , Na^+ , SO_4^{2+} , and Cl^- at 146.0 MPa (~ 100 km depth).

In the saturated eutectic ocean, there should be ample precipitate at the seafloor for the uptake of brine precipitate as the plume forms. If uptake is commensurate with the increase in solubility with temperature, enrichment leads to negative buoyancy at the seafloor, and the ocean will tend to stratify from the bottom up. In our calculations, we do not consider precipitation from the plume, but the effects of precipitation are discussed below.

In the more dilute ocean, enhanced salinity comes about through interaction with surrounding rock, perhaps via hydrothermal circulation (Kargel et al., 2000; McKinnon and Zolensky, 2003). We test a range of plume enrichment in a dilute MgSO_4 ocean from a calculation of the density anomaly, B [Eq. (1)].

2.3. Results

Density anomaly profiles are presented in Fig. 3 for 0.28 molal MgSO_4 , the minimum composition for which an unenriched plume ($n_{\text{plume}} = n_{\text{ocean}}$) is buoyant at all depths. The density anomaly versus european ocean depth is shown for three cases: (1) pure water plume and ocean, (2) saline

ocean and plume with the same composition, (3) plume with excess salinity in the same ocean. The density anomaly decreases as a plume rises, due to the pressure dependence of water's thermal expansion profile. Neutral buoyancy occurs where the density anomaly is zero.

The pure water and homogeneously saline systems have been treated (Collins et al., 2000; Thomson and Delaney, 2001; Goodman et al., 2004; Melosh et al., 2004). For pure water, the plume reaches neutral buoyancy at around 20 km depth; possibly well below the ice ceiling. In the case of homogeneous composition ($n_{\text{plume}} = n_{\text{ocean}}$), the positive temperature derivative of V^0 (illustrated in Fig. 4) makes the plume even less dense, moving the profile in the direction of greater buoyancy. As noted by Melosh et al. (2004), this causes the plume to remain buoyant up to the ice ceiling of Europa's ocean.

Saline enrichment of the plume, due to uptake at the seafloor, shifts the density anomaly profile in the direction of neutral buoyancy ($B = 0$). The magnitude of the density anomaly shift depends on both ambient and plume salinity, but the primary effect occurs over the entire range of ambient concentration of 0–1 molal MgSO_4 that we have considered. For the range of ocean salinities considered, the effect occurs for saline enrichment in the range 10^{-7} to 10^{-6} molal, ~ 3 orders of magnitude less than the increase in solubility commensurate with ΔT_{plume} . Critical rise heights are illustrated in Table 1 for a range of ocean salinities and enrichments, for $\Delta T_{\text{plume}} = 0.2$ mK and $\Delta T_{\text{plume}} = 1$ mK.

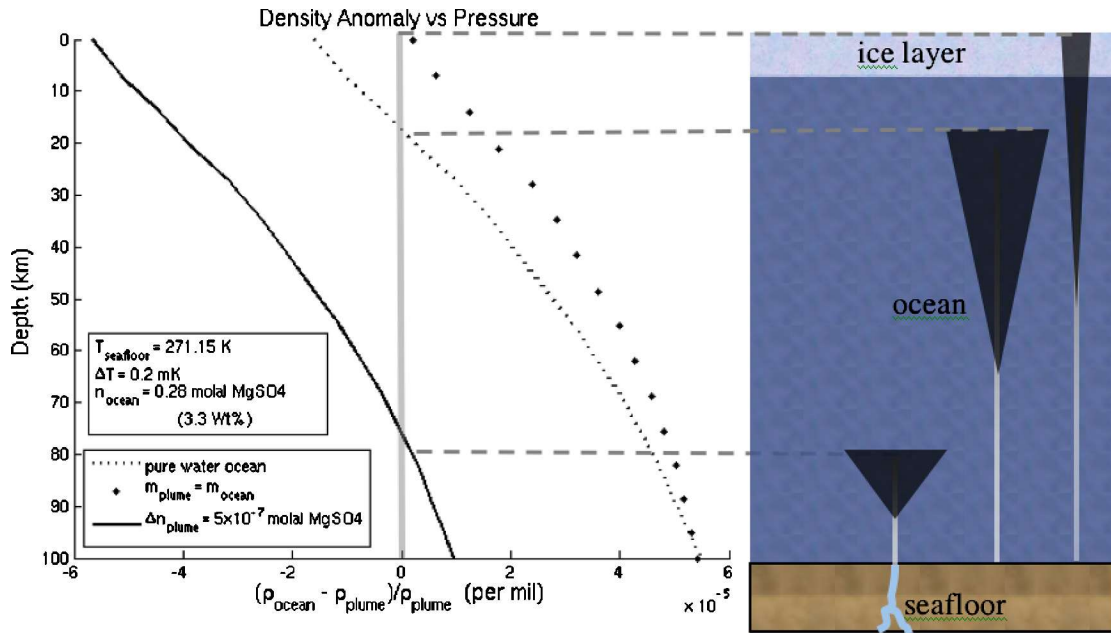


Fig. 3. Density anomaly in a european ocean for a plume with thermal anomaly of 0.2 mK. Ocean temperature is set to 271.15 K at the seafloor, with an adiabatic temperature decrease to ~ 270.85 K near the surface. A pure water plume in a pure water ocean (small dots) loses buoyancy around 20 km below the surface, where thermal expansion changes from positive below to negative above. A plume in a 0.28 molal MgSO_4 ocean (large dots) is less dense than the pure water plume with the same thermal anomaly, due to the increasing MgSO_4 molal volume with temperature (Fig. 4). The convergence of V^0 values at high pressure (Fig. 4) accounts for the convergence with the pure water plot at the seafloor. Adding salt to the plume (solid) via uptake at the seafloor increases the density anomaly of the plume, such that the plume loses buoyancy 20 km above the seafloor.

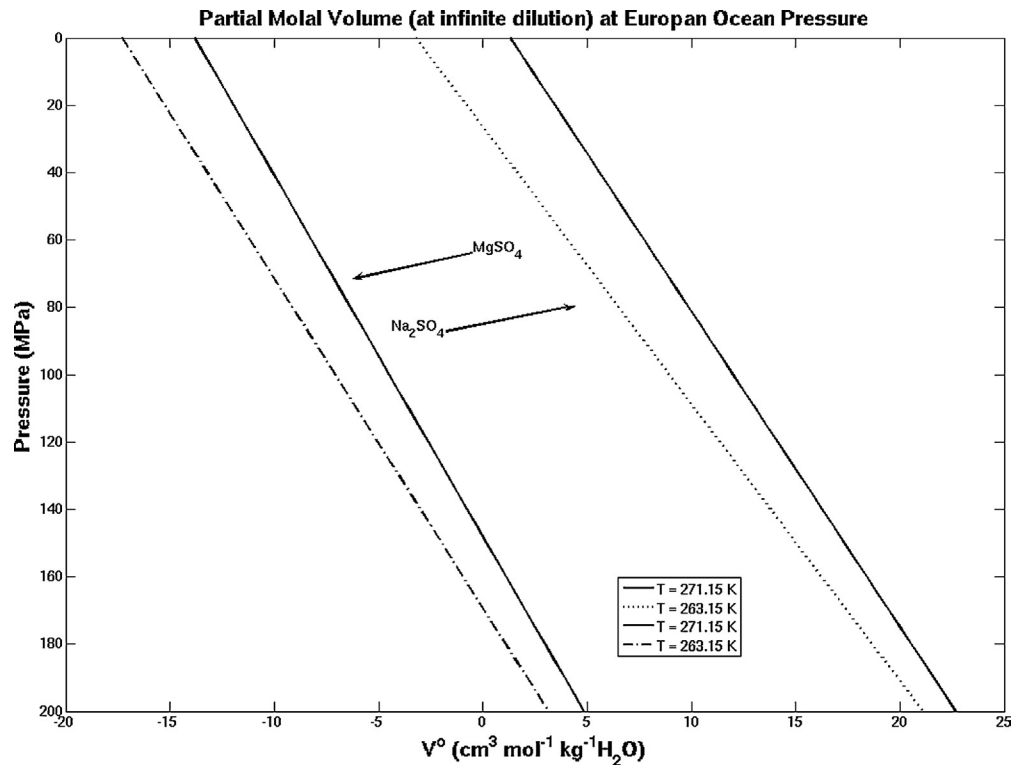


Fig. 4. Partial molal volumes for Na_2SO_4 and MgSO_4 , obtained using fits to [Millero \(1983\)](#) as suggested by [Marion et al. \(2005\)](#). V° increases with pressure and temperature.

Table 1

Depth of neutral buoyancy for a saline-enriched upwelling in Europa's ocean

T_{ocean} (K)	n_{ocean} (molal MgSO_4)	Δn (molal MgSO_4)	d_{neutral} (km)	
			$\Delta T = 0.2$ mK	$\Delta T = 1.0$ mK
271.15	0.28	5×10^{-7}	80	12
271.15	0.28	1×10^{-7}	12	5
271.15	0.28	6.5×10^{-7}	100	12
271.15	0.28	2×10^{-6}	100	60
271.15	1.00	1×10^{-6}	65	5
265.15	0.28	5×10^{-7}	80	27
265.15	1.00	1×10^{-6}	15	10

Note. The depth (below Europa's surface) of neutral buoyancy for 100 km ocean with average temperature 271.15 and 265.15 K. Loss of buoyancy at depth occurs for a small saline anomaly in a ring plume, over a range of 1–2 orders of magnitude. Increasing ambient salinity or plume thermal anomaly for a given ambient temperature decreases the depth of neutral buoyancy. Lowering ocean temperature appears to increase the depth of neutral buoyancy, though it is difficult to discern a change for the smaller thermal anomaly.

Whether plume buoyancy can be maintained over the entire depth of Europa's ocean depends on the ocean's depth, temperature profile, and composition. The degree to which a plume at the seafloor acquires excess salinity, and thereby the height to which its rise is limited, depends on the change in solubility and dissolution rate with temperature, the existence of a reservoir of salt to draw from, the dimensions of the reservoir, and the residence time of the plume at the seafloor.

2.4. Double-diffusive convection in Europa's ocean

Loss of plume buoyancy at a height determined by saline enrichment suggests that Europa's ocean is stratified to some

degree. This contrasts with the assertion ([Thomson and Delaney, 2001](#); [Goodman et al., 2004](#)) that the ocean has a large Rayleigh number, and convective heat and fluid transport are ubiquitous. Here, we explore the possibility of convection in a lower layer that is warm and saline, with convection also occurring above, in a cooler, more dilute layer, with a diffusive boundary between them.

Generally, the stability of a non-convecting, or stagnant, layer will be determined by its ability to transport heat to the overlying water, either by advection or diffusion. If there is a gradient in both temperature and salinity, double-diffusive convection can occur, in which the magnitudes of the two gradients determine the style of heat and material transport.

In the simplification that density scales linearly with temperature and salinity (Turner, 1973), one arrives at two non-dimensional numbers that characterize the double-diffusive system:

$$Ra = \frac{g\alpha\Delta T d^3}{\kappa\nu}, \quad (6)$$

$$Rs = R_\rho Ra = \frac{g\beta\Delta S d^3}{\kappa\nu}, \quad (7)$$

$$R_\rho \equiv \frac{\beta\partial S/\partial z}{\alpha\partial T/\partial z}. \quad (8)$$

Ra is the Rayleigh number and convection occurs for $Ra > O(10^3)$. d is the characteristic scale length for the system (here, the height of Europa's ocean). Rs is the salinity Rayleigh number, in which $\beta = \partial \ln \rho / \partial S$ is the coefficient of saline contraction, analogous to α in Ra .

When $1 \leq R_\rho < 10$ a double-diffusive system is observed (Fernando, 1989), in which a well-mixed (convecting) lower layer is separated from a well-mixed upper layer, with a step profile in salinity and temperature between them. The better-studied regime of salt fingering (Merryfield, 2000) occurs when salinity and temperature decrease with depth. In this situation, saltier fluid from above enters the lower, more dilute layer, and begins to rapidly descend in characteristic "finger" plumes. The subtler, less studied, diffusive regime, in which heat and salt diffuse upward, occurs when salinity and temperature increase with depth, with $1 \leq R_\rho < 10$, and has been observed in cold and isolated systems on Earth (Kelley et al., 2003; see below).

2.4.1. Feasibility of double-diffusive convection in Europa's ocean

Here, we assess the feasibility of double-diffusion in Europa's ocean based on the above discussion, assuming the range of relevant R_ρ is the same. We assume properties of water as mentioned in previous sections. R_ρ is defined in Eq. (7). For the aqueous MgSO_4 system at 271.15 K (Hogenboom et al., 1995), the coefficient of saline contraction is $\beta \sim 0.1 \text{ mol}^{-1}$. For a range of difference in potential temperature between the top and bottom of Europa's ocean, $\Delta T = 10^{-4}$ to 10^1 K, the salinity change over the entire ocean to attain $1 \leq R_\rho < 10$ is: $\Delta S \sim 10^0$ – 10^{-6} . ΔS over the entire 100 km ocean depth is 0 to 7 orders of magnitude greater than the salinity difference chosen to illustrate the stagnation (Fig. 3; Table 1).

If double-diffusive convection occurs in an ocean near saturated composition, precipitation from a rising plume would cause salt to "rain down" into the ocean below. Rain out would enhance the salinity of the ocean below while transferring heat upward rather than keeping it at the level of neutral buoyancy. The resulting disturbance to the relative thermal and saline gradients would increase R_ρ above the range where double-diffusive convection is observed in laboratory experiments.

2.4.2. Double-diffusive convection in Earth's waters

The existence of stable double-diffusion systems in Earth's waters has been inferred from the step, or staircase, nature in salinity profiles in the Weddel Sea, in the Greenland Sea, below ice island T-3 in the Arctic (all so far mentioned: Kelley et al., 2003) in ice-covered Lake Vanda in Antarctica (Hoare, 1966), and in the Black Sea (Kelley et al., 2003; Özsoy and Ünlüata, 1997).

In the Black Sea, where temperature and salinity increase with depth and the bottom is heated geothermally, the bottom convecting layer is stable and nearly isothermal, 300–400 m thick, $O(1000)$ years old, and perhaps 0.5 K warmer and 3 wt% more saline than the overlying water (Özsoy and Ünlüata, 1997).

Double diffusive convection is observed directly above (and possibly below) hydrothermal systems in the Red Sea (Blanc and Anschutz, 1995; Anschutz et al., 1998). Convective layers in the Red Sea moved upward in the thirty years since they were first observed in the 1960s (Anschutz et al., 1998). No such change has been noted for Lake Vanda.

Kelley et al. (2003) speculate that high-latitude regions on Earth may be largely susceptible to diffusive regime double-diffusive convection, based on analysis of R_ρ in areas where salinity and temperature profiles have been measured.

3. Implications for Europa's geologic history

Layer evolution could be indicated by the relatively recent occurrence of melt-through features in Europa's surface history. Surface stratigraphy of the icy lithosphere (Figueredo and Greeley, 2004) suggests melt-through rather than tectonic processes dominate Europa's surface dynamics in the current epoch. If we take this to indicate a trend toward complete resurfacing of the ice shell, the spread of a warm salty layer to the surface may be key to explaining the time scale of the process. If the warm layer moves upward with the addition of heat and salt, as in the Red Sea (Anschutz et al., 1998), chaos features could indicate intrusion of a warm plume after the layer reaches some critical height.

Might storage of heat be responsible for catastrophic melt-through and resurfacing of Europa's ice? The minimum ocean temperature above freezing needed to melt the entire ice shell can be estimated by solving for ΔT_{ocean} in

$$M_{\text{ocean}} C_p \Delta T_{\text{ocean}} = M_{\text{ice}} Q_{\text{ice}} + M_{\text{ice}} C_p \Delta T_{\text{ice}}, \quad (9)$$

the equation of heat transferred from the ocean in order to melt the entire volume of the ice shell. ΔT_{ocean} is the amount, in K, by which the ocean temperature exceeds the freezing temperature of the ice shell. Q_{ice} is the latent heat of melting, 334 kJ/kg. M_{ocean} and M_{ice} are the masses of the ocean and ice layers $M \sim (4/3)\pi(R_{\text{top}}^3 - R_{\text{bottom}}^3)\rho_{\text{water}}$, respectively. ΔT_{ice} is the amount, in K, by which the ice shell must be warmed in order to bring it to the melting temperature. We take ΔT_{ice} to be ~ 86 K, the average difference be-

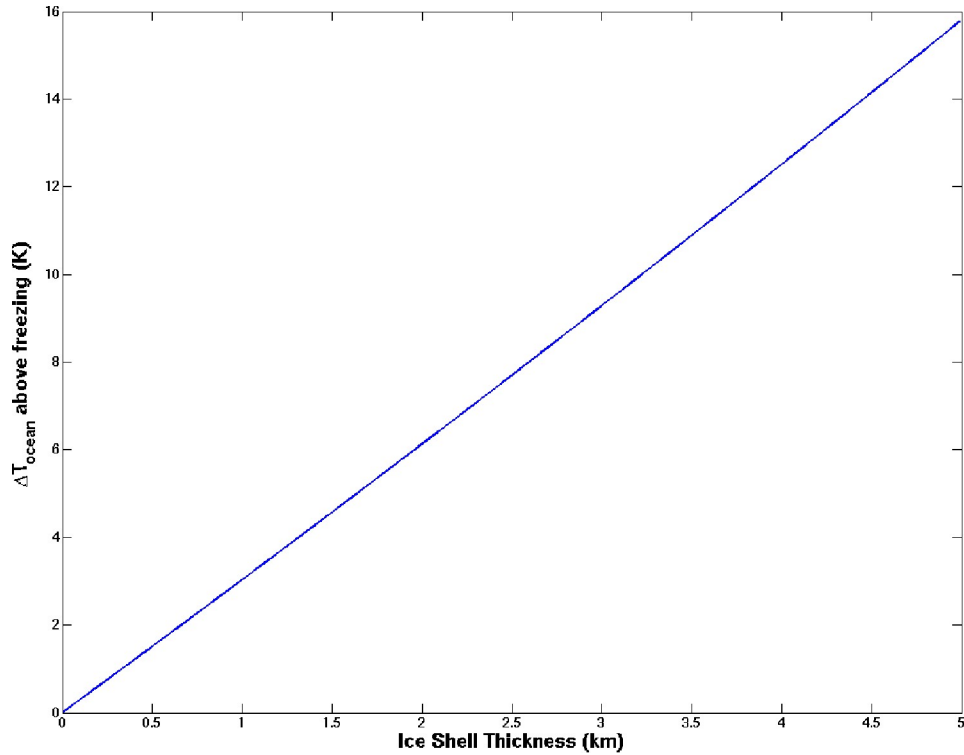


Fig. 5. Ocean temperature, above melting temperature of the ice shell, needed to melt Europa's ice shell, as a function of ice shell thickness. Europa's H₂O layer is assumed to be 100 km thick.

tween Europa's surface temperature ($T_{\text{surface}} \sim 100$ K) and an ocean ceiling at 273 K. With C_p of 4 kJ/kg, the ocean temperature above freezing needed to melt a 2 km ice shell is around 5 K. Fig. 5 illustrates how the ocean temperature above freezing scales with ice thickness.

The time scale to heat the ocean by $O(1)$ K can be estimated as

$$t \sim \frac{M_{\text{ocean}} C_p \Delta T}{F} \quad (10)$$

$F = 4\pi R_{\text{seafloor}}^2 H$ is the total heat flux from the seafloor, in W. The time scale to heat a 100 km ocean by $\Delta T = O(1)$ K is $t \sim 100$ –10 Myr for the lower and upper estimates of heat inputs used in this paper ($H = 0.15$ to 4 mW m^{-2}), respectively (for other assumed values, see Appendix A). Thus, resurfacing of the ice shell by heat stored in Europa's ocean is possible on the time scale of Europa's 30–80 Myr surface age (Zahnle et al., 2003) if the ice shell is thin (< 1 km) or heat storage is effective enough to heat the ocean to $O(1)$ K above the melting temperature of the ice shell. Neither of these possibilities is ruled out by current knowledge of Europa or studies of double-diffusive convection to date.

Upward progress of convecting layers is not the only possibility. For instance, if tidal and radiogenic heat inputs are dissipated effectively, perhaps through a thin ice shell and thin diffusive layers separating convecting zones, double-diffusive layers could actually move downward. Future work should examine heat balance in Europa's ocean.

4. Conclusion

We have demonstrated the possible existence of a layered system analogous to double-diffusive convection systems on Earth. Saline enrichment in nascent upwellings provides a depth-dependent stay on buoyancy similar to the effect pointed out by Melosh et al. (2004). The effect implies stratification of the ocean, and may indicate a means for the onset of layered convection. As a test of the feasibility of double-diffusive convection, we find mild saline stratification consistent with parameters expected for layering systems (Kelley et al., 2003).

Heat storage and release via stratification, mediated perhaps by double-diffusive convection, may be the temporal link between surface and subsurface processes, only if the ice shell is very thin (< 1 km) or the mean ocean temperature significantly higher (> 1 K) than the freezing temperature of the ice shell. Further quantitative treatment of compositional effects in Europa's ocean is necessary, pending the contribution of more experimental data on sulfate salts, and investigation of double-diffusive phenomena, both in the lab and via exploration of systems in Earth's waters.

Acknowledgments

This project was supported by the National Science Foundation's IGERT program, grant number: DGE-9870713,

“IGERT: Astrobiology: Life in and beyond Earth's Solar System,” the NASA Astrobiology Institute, and National Science Foundation award EAR-0337796. We also thank Giles Marion for helpful discussion, and Jason Goodman and Constantine Thomas for their thoughtful reviews.

Appendix A. Revised heat flow estimates

An estimate of ice thickness should be consistent with expected surface and bottom ice temperatures, at least in the simplification that the ice shell is not heated internally by tidal forces and cooling of the ice shell is entirely diffusive. Heat is transported by diffusion through the european ice shell. That is,

$$\frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial z^2} \quad (\text{A.1})$$

where z denotes depth and κ is the thermal diffusivity of water ($\sim 1 \times 10^{-7} \text{ m}^2 \text{ s}^{-1}$). Thus, the diffusion time scale is $t \sim d^2/\kappa$, on the order of Myr for a $O(\text{km})$ thick ice shell. Heating in this region is determined by conduction (diffusion of heat):

$$Ht = \rho C_p d \Delta T, \quad (\text{A.2})$$

where d is the depth of the diffusive region and H is the heat production in W m^{-2} . Goodman et al. (2004) consider a broad range of global heat input to the base of Europa's ocean, for the range of extremes in which all or none of the tidal dissipation occurs in the rock underneath the ocean, at depth R : $F = 4\pi R^2 H = 0.1\text{--}100 \text{ GW}$, or H around $0.04\text{--}4 \text{ mW m}^{-2}$, bracketing the estimates of O'Brien et al. (2002) and Thomson and Delaney (2001).

The difference in temperature between the top of the ice ($T_{\text{surface}} \sim 100 \text{ K}$) and the bottom (melting temperature) is $\Delta T \sim 170 \text{ K}$. For the lower heat flow estimate $0.04 \text{ (mW m}^{-2}\text{)}$, the ice thickness needed for a liquid base is 200 km , twice our assumed ocean depth, though only slightly thicker than H_2O layer thickness allowed by Galileo moment of inertia measurements (Anderson et al., 1998). For the upper heat flow estimate, a self-consistent ice thickness is 4 km . The smallest heat flow that leads to melting ($\Delta T \sim 170 \text{ K}$) within 100 km of Europa's surface is $H \sim 0.15 \text{ mW m}^{-2}$. Therefore, we consider $H \sim 0.15$ to 4 mW m^{-2} .

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